

Software Requirements Specification (SRS) Template

Maternawell Nigeria

Prepared by Chigozie Emmanuel Ndubuaku

AFRICAN LEADERSHIP COLLEGE OF HIGHER EDUCATION

31 July 2026

Description of the concept of the Mission.....	1
Gap Analysis: What Already Exists and Why It Isn't Enough.....	2
The Hypothesis of the Solution.....	3
References.....	4
1. Introduction.....	9
1.1 Purpose.....	9
1.2 Document Conventions.....	9
1.3 Intended Audience and Reading Suggestions.....	10
1.4 Product Scope.....	10
2. Overall Description.....	10
2.1 Product Perspective.....	10
2.2 Product Functions.....	10
2.3 User Classes and Characteristics.....	11
2.4 Operating Environment.....	11
2.5 Design and Implementation Constraints.....	11
2.6 User Documentation.....	11
2.7 Assumptions and Dependencies.....	12
3. External Interface Requirements.....	12
3.1 User Interfaces.....	12
3.2 Hardware Interfaces.....	12
3.3 Software Interfaces.....	12
3.4 Communications Interfaces.....	12
4. Requirement Specification	
4.1 Digital EPDS Screening.....	13
4.2 Automated Risk Scoring and Classification.....	13
4.3 Referral Recommendation Engine.....	13
4.4 Offline-First Data Capture and Sync.....	14
4.5 Referral Follow-Up Tracking.....	14
4.6 Anonymous Self-Referral.....	14
4.7 Health Worker Authentication and Facility Case Dashboard.....	15
5. Other Nonfunctional Requirements.....	15
5.1 Performance Requirements.....	15
5.2 Safety Requirements.....	16
5.3 Security Requirements.....	16
5.4 Software Quality Attributes.....	16
5.5 Business Rules.....	17
6. Appendix.....	18
Appendix A: Glossary.....	18
Appendix B: Analysis Models.....	18

Revision History

Name	Date	Reason For Changes	Version
Ndubuaku Chigozie Emmanuel	July 15, 2026	Initial draft: proposal and SRS based on federated-learning concept, later replaced.	1.0
Ndubuaku Chigozie Emmanuel	July 16, 2026	Full revision: scoped to Maternawell Nigeria (postpartum depression screening); independent research base with verified citations, gap analysis, and legal grounding added.	1.0
Ndubuaku Chigozie Emmanuel	July 31, 2026	Added the UML diagrams in Appendix B	2.0

1. Introduction

1.1 Purpose

This document outlines both functional and non-functional requirements for Maternawell Nigeria, a software program to empower primary health center (PHC) operators to screen maternal postnatal depression utilizing a research tool validated in Nigeria, as well as refer patients diagnosed with those at risk to a support institution. Essentially, this represents an enhancement of the task-shifting care model used in the MeHPriC program implemented in Lagos State (Adewuya et al., 2025).

1.2 Document Conventions

The system's features refer to numbers 4.1, 4.2, etc. with their corresponding priority assigned to each feature and functional requirements denoted by their FR-# numbers. Non-functional requirements appear in section 5.1-5.5 as NFR-#. The word "must" is used to symbolize mandatory requirements and "should" to indicate high priority, but one that does not block the release of the feature.

1.3 Intended Audience and Reading Suggestions

The intended readership includes student developer(s), those teaching the course who will evaluate this proposal, and any organization or PHC personnel who consulted during the pilot phase of development. New readers should consult Section 2 before proceeding to the requirements outlined in Sections 4 and 5.

1.4 Product Scope

Maternawell Nigeria automates the process of EPDS in creating a workflow for PHC workers which entails scoring such against validated cutoffs in Nigeria, classifying risk levels, and recommending referral actions based on MeHPriC's stepped-care framework. The system is accessible offline and synchronizes when connected. Not included in the current scope of

activities are clinical diagnoses (the solution is meant for screening purposes) as well as prescription and treatment management along with national EMR integration.

2. Overall Description

2.1 Product Perspective

Maternawell Nigeria represents a new unified system. It is being described as a digital extension of the previously verified task shifting approach from MeHPriC and not a substitute for it. The goal is to enable the use of MeHPriC-style organized screening in places where qualified supervisors trained in the mhGAP approach are not available..

2.2 Product Functions

- Provide a digital EPDS assessment to a new mother.
- Electrically assess the result of the screening based on the Nigerian endorsed point of demarcation and define the hazard tier.
- Advise a referral activity based on the hazard level.
- Collect and preserve details offline until chances for connection arise.
- Record the referral outcome.
- Facilitate anonymous self-referral process.

2.3 User Classes and Characteristics

Primary Health Center Workers are the main users of the app; they do screening tests and have moderate technical skills, but little training in digital technology and this goes along with MeHPriC training model results that show that structured tools have increased efficacy of 69.1% of users (Adewuya et al. 2025). Facility Supervisors have limited usage; they can view the dashboard of cases and their follow-up. Mothers use the app indirectly through the anonymous self-referral process without having to upload their personal information.

2.4 Operating Environment

The Android app (8.0+) runs on reasonably-priced devices such as low-to-mid-range smartphones and tablets, knowing that many people will use it without a connection, as only 23% of rural areas in Nigeria have internet access.

2.5 Design and Implementation Constraints

They have to follow Section 30 of the Nigeria Data Protection Act, 2023, which describes health data as sensitive personal data involving obtaining explicit consent and higher levels of protection. It should be user-friendly for personnel with basic digital skills since the onboarding time will be short. Reliability of electricity supply and connectivity should not be assumed. The clinical scoring should adhere to the validated EPDS cutoff in Nigeria (≥ 9 ; Uwakwe & Okonkwo, 2003) rather than following a customized or imported Western cutoff.

2.6 User Documentation

The following documentation will be delivered alongside the software:

- **In-app onboarding walkthrough:** a guided, tap-through sequence shown on first login that walks a health worker through completing one full screening, triggered automatically the first time they log in.
- **One-page printed quick-reference guide:** laminated and posted near the PHC workstation, covering the EPDS scoring cutoff (≥ 9), the Item-10 escalation rule, and what each risk tier means in plain language, for moments when a worker doesn't want to dig through the app mid-consultation.
- **In-app contextual help:** a small "?" icon on each screen offering a one-line explanation of that step, rather than a separate manual a worker has to go find.
- **Supervisor guide:** a short separate document (2-3 pages) covering how to use the facility case dashboard, review escalated cases, and interpret follow-up status, since supervisors need different information than the workers doing the screening.
- **Yoruba-language reference materials:** for South-West deployments, a translated version of the quick-reference guide aligned with the Yoruba-validated EPDS instrument (Adewuya et al., 2006).
- **FAQ / troubleshooting sheet:** covering common issues specific to the offline-first design, e.g., "what happens if I lose connectivity mid-screening," "how do I know if my data synced."

2.7 Assumptions and Dependencies

The possibility of an Android platform presence in at least one of the facilities attended by PCH staff should be acknowledged, as well as the potential acceptance of Yoruba-translated EPDS (Adewuya et al., 2006) for use in South Western operations (the requirements for translation validation in other regions are not within the ambit of this edition). The success of the exploratory study may depend on learning institutions and authorities cooperating with the implementing organization.

3. External Interface Requirements

3.1 User Interfaces

The mobile interface should be simple with large touch targets featuring the login screen, the screening form (with only one question), the results/risk-tier screen, referral recommendation screen and the supervisor's case dashboard. The interface should be in English by default and contain labels in Yoruba if needed.

3.2 Hardware Interfaces

Runs on Android smartphones/tablets with local on-device storage. No specialized hardware required for this version.

3.3 Software Interfaces

Local on-device database (e.g., SQLite) for offline storage; a lightweight backend/sync service for centralizing records once online. No third-party EMR integration in this version.

3.4 Communications Interfaces

Sync communications use HTTPS when connectivity is available. No real-time communication is required for core screening functionality.

4. Requirement Specification

4.1 Digital EPDS Screening

Summary: Provides health personnel with the Edinburgh Postnatal Depression Scale in digital, simplified format.

- Important Aspect: Essential
- Technical Specifications:
- **FR-1:** The system is able to deliver the Edinburgh Postnatal Depression Scale in digital format easily and quickly.
- **FR-2:** The system will present a single question on each screen and there will be minimum difficulty in answering the questions.

4.2 Automated Risk Scoring and Classification

Summary: Assess every screening exercise that has been finished and categorize each through the use of the Nigerian-adopted EPDS cutoff score, as opposed to an arbitrary cut-off score or imported cutoff score.

- **Need:** Essential
- Functional Specifications:
- **FS-3:** The program will compute the EPDS total score automatically once the form has been submitted.
- **FS-4:** The program will categorize each score into a risk classification (low/moderate/high) by means of the Nigerian-validated cutoff score of ≥ 9 (Uwakwe & Okonkwo, 2003).
- **FS-5:** The program will designate any non-zero score obtained for EPDS Item 10 (self-harm ideation) as a serious case that needs to be expedited the same day, irrespective of the overall score.

4.3 Referral Recommendation Engine

Summary: Proposes a particular action to be taken for every risk category, based on the successful MeHPriC approach, thus defining not only the issue but giving useful instructions to the healthcare practitioner on what steps to take.

Importance: Essential requirement

Functional specifications:

- **FR-6:** The software must suggest a referral for each risk category (i.e., community/peer support, counseling at a primary health care facility, psychiatric referral required immediately).
- **FR-7:** The software must show the recommendation immediately after assessing the risk level without needing further assistance from health practitioners.

4.4 Offline-First Data Capture and Sync

Summary: Guarantees the operation of the system with no connection, as only 23% of communities in rural Nigeria have internet access right now (NCC, 2025), while the sync occurs automatically once the connection becomes available.

Priority: Must Have

Functional Requirements:

- **FR-8:** The system needs to save completed screenings to the mobile device in offline mode.
- **FR-9:** The system should transfer the information saved locally to the central log without user input after the connection is available.

4.5 Referral Follow-Up Tracking

Summary: Enables the health professionals and their supervisors monitoring the results of the referral closing the gap that is currently left by the improvised paper-based screening.

Priority: Should Have

Functional Requirements:

- **FR-10:** The system shall facilitate the updating follow-up status of a referral by the health workers (contacted / pending / completed).
- **FR-11:** The system must indicate the follow-up status at the facility case dashboard (see Feature 4.7).

4.6 Anonymous Self-Referral

Summary: Enables mothers to do screenings and seek help without mentioning their symptoms directly, thereby overcoming a stigma issue highlighted in PPD literature on Nigeria (Atuhaire et al., 2020).

Priority: Required

Requirements:

- FR-12: The system must enable mothers to follow a self-screening procedure anonymously, without involving a health worker in the process.
- FR-13: The system must channel self-referrals made anonymously through the same structured referral system.

4.7 Health Worker Authentication and Facility Case Dashboard

Description: Restricts data entry to authorized staff and gives facility supervisors visibility into active and pending cases across the facility.

Priority: Should Have

Functional Requirements:

- FR-14: The system must require health workers to log in with a unique credential before accessing or entering screening data.
- FR-15: The system must display a dashboard of a facility's active and pending referral cases to designated supervising staff.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Req ID	Description
NFR-1	Risk scoring must return a result in under 1 second on low-to-mid-range Android devices.
NFR-2	Sync operations must complete successfully over minimal (2G-equivalent) bandwidth, reflecting median rural network conditions (NCC, 2025).

NFR-3	The system must remain usable for a full clinic shift (approx. 8-12 hours) without crashing or losing unsaved data.
-------	---

5.2 Safety Requirements

Req ID	Description
NFR-4	Any Item-10 (self-harm ideation) response must trigger a same-day escalation flag that cannot be dismissed without supervisor acknowledgment (see FR-5).
NFR-5	The system must never present itself as providing a clinical diagnosis; all screens must clearly label output as a screening result requiring appropriate follow-up, not a diagnosis.

5.3 Security Requirements

Req ID	Description
NFR-6	Screening data is classified as sensitive personal data under Section 30 of the Nigeria Data Protection Act, 2023, and must be processed only with explicit consent and a higher standard of technical safeguard (Lexworth Legal Partners, 2023).
NFR-7	The system must log all screening and referral actions with a timestamp and health-worker ID for audit and NDPA accountability purposes.
NFR-8	Locally stored data on-device must be encrypted at rest.

5.4 Software Quality Attributes

Attribute	Description
Usability	A health worker with minimal digital literacy training must be able to complete a screening unassisted after a single onboarding session, consistent with task-shifting outcomes observed under MeHPriC's mhGAP training model (Adewuya et al., 2025).
Reliability	The system must remain fully functional for screening, scoring, and local storage with zero internet connectivity.

Portability	The system must run on low-cost Android devices (Android 8.0+), the dominant device class among Nigeria's roughly 150 million active mobile connections (DataReportal, 2025).
Scalability	The system's facility-level architecture must be replicable across PHCs without redesign, modeled on MeHPriC's demonstrated scale-up across 57 flagship facilities and 852 trained workers (Adewuya et al., 2025).
Cultural Sensitivity	Screening language and referral prompts must be reviewed for stigma-sensitivity, and must support the Yoruba-validated EPDS translation for South-West Nigerian deployments (Adewuya et al., 2006).

5.5 Business Rules

- Only authenticated health workers may enter or view screening data for named (non-anonymous) mothers.
- Facility supervisors may view aggregated case status across their facility but not raw EPDS item-level responses, unless the case has been escalated.
- Any Item-10 self-harm flag must be visible to a facility supervisor within the same clinic day, regardless of the health worker's own follow-up actions.
- Anonymous self-referral records are not linked to any identifiable mother record unless she voluntarily provides contact information during the referral step.

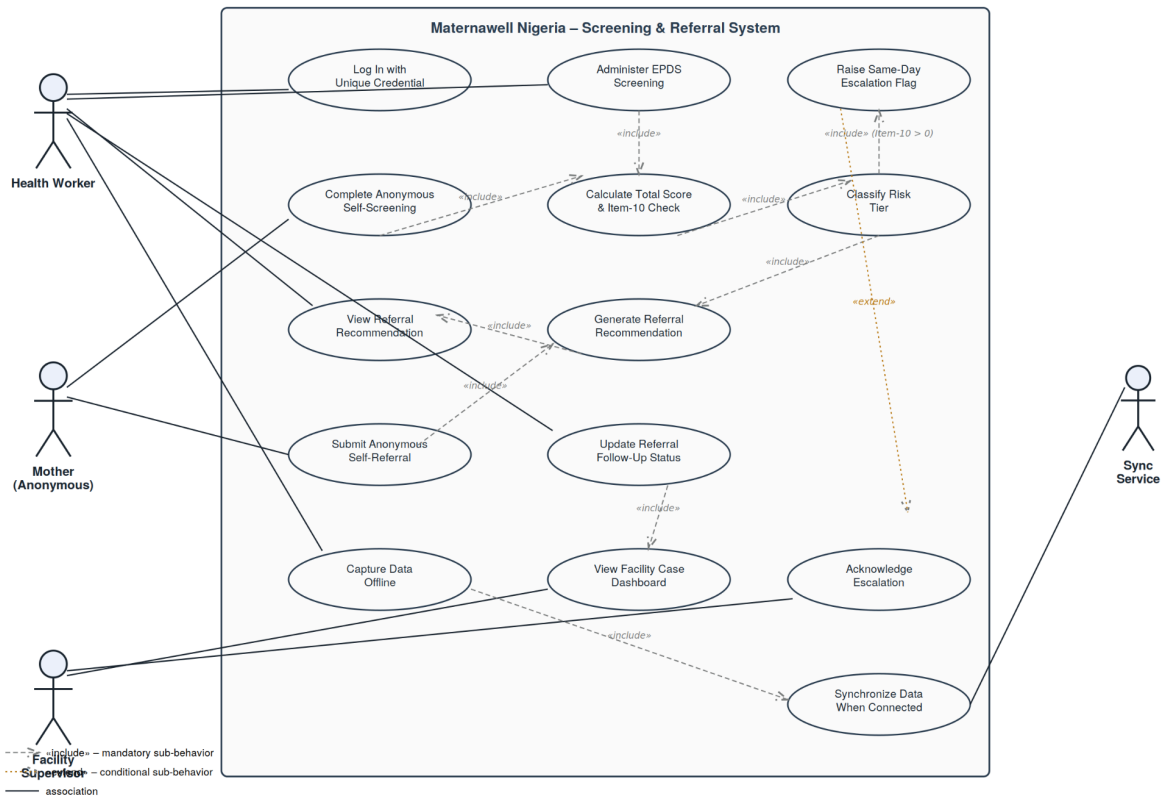
6. Appendix

Appendix A: Glossary

- EPDS: Edinburgh Postnatal Depression Scale, a validated 10-item clinical screening questionnaire (Cox et al., 1987).
- PPD: Postpartum Depression.
- PHC: Primary Health Center.
- NDPA: Nigeria Data Protection Act, 2023.
- mhGAP: WHO's Mental Health Gap Action Programme, the clinical framework underlying MeHPriC's task-shifted care model.
- MeHPriC: Mental Health in Primary Care, the Lagos State program this project's referral logic is modeled on.

- Offline-first: a design approach where core functionality works without internet, syncing only when connectivity becomes available.

Appendix B: Analysis Models



UseCase Diagram:

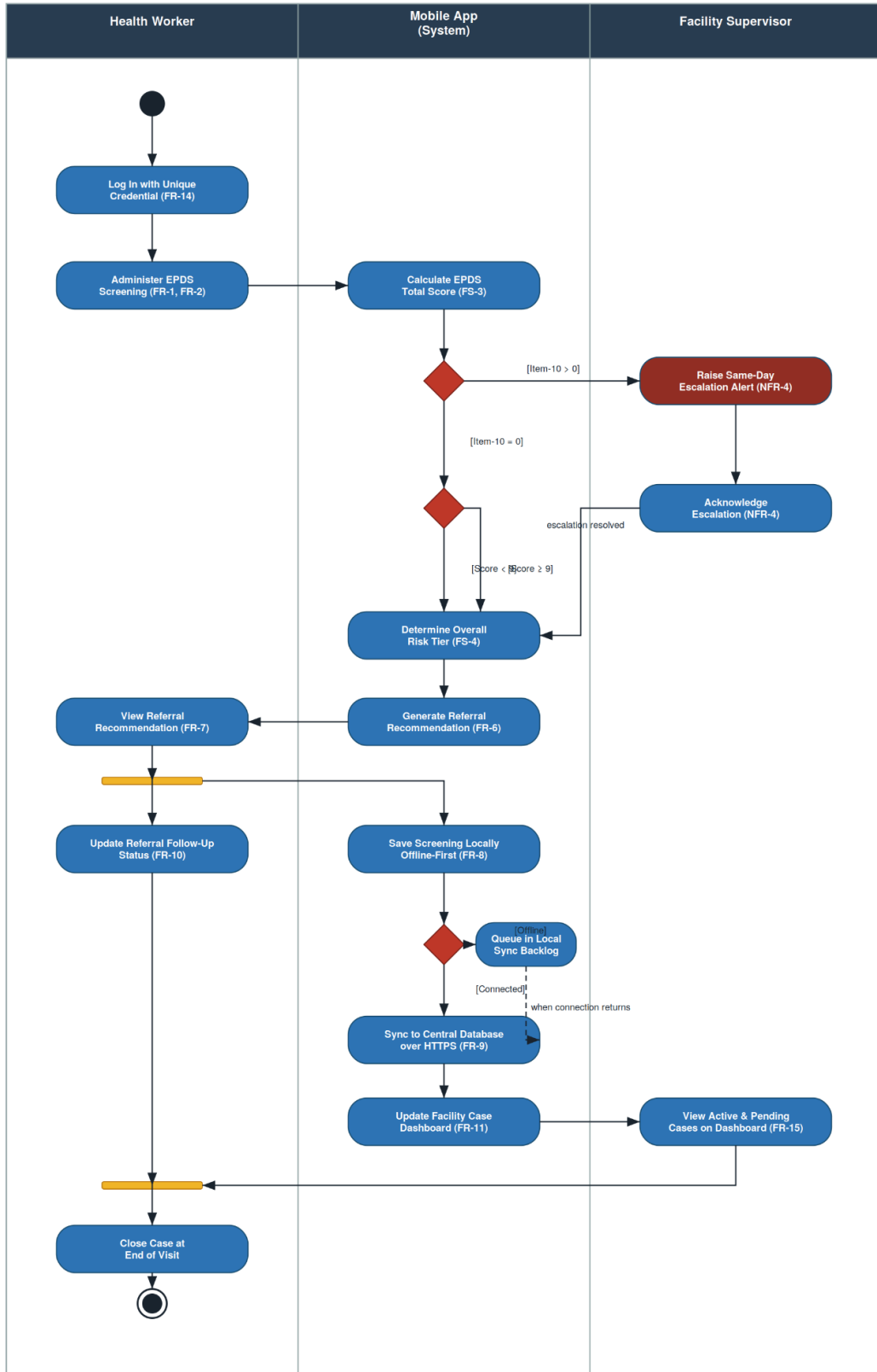
This diagram shows who uses my system and what each person can do. My actors are the Health Worker, the Mother, the Facility Supervisor, and the Sync Service, which is a secondary actor since it represents a background process rather than a person. Administer Screening and Complete Anonymous Self-Screening both have an include relationship with Calculate Score, since scoring is a mandatory step in both cases. Calculate Score has an include relationship with Classify Risk Tier, which itself has a conditional include with Raise Escalation Flag, restricted to the case where Item-10 is greater than zero. I used an extended relationship for Acknowledge Escalation, since it is optional behavior that only triggers when an escalation actually fires, rather than mandatory behavior that runs every time like an include. Classify Risk

```
classDiagram
    class User {
        -userId : String
        -fullName : String
        -role : UserRole
        -facilityId : String
        -passwordHash : String
        +login(credential) : Session
        +logout() : void
    }
    class HealthWorker {
        -trainingCompleted : Boolean
        +administerScreening(mother) : Screening
        +updateFollowUp(referralId, status) : void
    }
    class RiskClassifier {
        -nigeriaCutoff : Integer = 9
        +classify(screening) : RiskTier
        +checkItem10(escalation(screening)) : Boolean
    }
    class Facility {
        -facilityId : String
        -name : String
        -region : String
        -state : String
        +getActiveCases() : List~Screening~
    }
    class FacilitySupervisor {
        -supervisedFacilityId : String
        +viewCaseDashboard() : List~CaseSummary~
        +acknowledgeEscalation(flagId) : void
    }
    class Mother {
        -referenceId : String
        -isAnonymous : Boolean
        -contactInfo : String
        +startSelfScreening() : Screening
    }
    class Screening {
        -screeningId : String
        -facilityId : String
        -healthWorkerId : String
        -motherReferenceId : String
        -responses : EPDResponse[10]
        -totalScore : Integer
        -item10Score : Integer
        -channel : ScreeningChannel
        -createdAt : DateTime
        -syncStatus : SyncStatus
        +submit() : void
        +calculateTotalScore() : Integer
        +isItem10Positive() : Boolean
    }
    class ReferralRecommendation {
        -referralId : String
        -screeningId : String
        -riskTier : RiskTier
        -recommendedAction : String
        -generatedAt : DateTime
        +generate(riskTier) : ReferralRecommendation
    }
    class ReferralFollowUp {
        -referralId : String
        -status : FollowUpStatus
        -updatedAt : DateTime
        -notes : String
        +updateStatus(status) : void
    }
    class AuditLogEntry {
        -logId : String
        -actorId : String
        -action : String
        -timestamp : DateTime
    }
    class EscalationFlag {
        -flagId : String
        -screeningId : String
        -triggeredAt : DateTime
        -acknowledgedBy : String
        -acknowledgedAt : DateTime
        -status : EscalationStatus
        +acknowledge(supervisorId) : void
    }
    class EPDResponse {
        -itemNumber : Integer (1..10)
        -questionText : String
        -selectedValue : Integer (0..3)
    }
    class SyncManager {
        -pendingQueue : List~Screening~
        -lastSyncedAt : DateTime
        +queueForSync(screening) : void
        +syncWhenConnected() : void
        +isConnected() : Boolean
    }
    class RiskTier {
        <<enumeration>>
        Low
        Moderate
        High
    }
    class FollowUpStatus {
        <<enumeration>>
        Pending
        Contacted
        Completed
    }
    class SyncStatus {
        <<enumeration>>
        Unsynced
        Queued
        Synced
    }

    User --> HealthWorker
    User --> Facility
    User --> FacilitySupervisor
    User --> Mother
    HealthWorker --> Screening : administers
    HealthWorker --> ReferralRecommendation : generates
    HealthWorker --> ReferralFollowUp : updates
    HealthWorker --> RiskClassifier : produces
    RiskClassifier --> Screening : evaluates
    RiskClassifier --> EscalationFlag : produces
    Facility --> Screening : recorded at
    FacilitySupervisor --> EscalationFlag : acknowledges
    Mother --> Screening : undergoes
    Mother --> ReferralRecommendation : based on
    Screening --> ReferralRecommendation : triggers
    Screening --> ReferralFollowUp : based on
    Screening --> EPDResponse : composed of
    Screening --> SyncManager : queued by
    ReferralRecommendation --> ReferralFollowUp : tracked by
    ReferralFollowUp --> AuditLogEntry : generates
    ReferralFollowUp --> FollowUpStatus : has
    SyncManager --> SyncStatus : sets
    EscalationFlag --> FacilitySupervisor : evaluates
    EscalationFlag --> Screening : recorded at
```

Class Diagram:

I made this diagram to show the main objects in my system and how they relate to each other. HealthWorker, FacilitySupervisor and SystemAdmin form a generalization hierarchy, where each subclass inherits the attributes and methods of the class below it. Screening is the central entity. It holds the score, the Item-10 flag, the risk tier and the sync status, and it has a composition relationship with EPDSResponse, since a screening is made up of exactly ten responses that cannot exist without it. Mother and HealthWorker each have an association with Screening, labeled undergoes and administers. Once a Screening is scored, it has an association with ReferralRecommendation, which in turn has an aggregation relationship with ReferralFollowUp, since a follow-up record can be tracked independently of the recommendation that created it. If Item-10 is positive, the Screening has a dependency relationship with EscalationFlag, meaning the flag is generated as a result of the screening logic rather than being a permanent part of it. FacilitySupervisor has an association with EscalationFlag through the acknowledged operation. I also added a SyncManager with a dependency on Screening for offline queuing, an AuditLogEntry, and a FacilityManager that has an aggregation relationship with Facility. RiskTier, FollowUpStatus and SyncStatus are enumerations, and their owning classes have a dependency (a «uses» relationship) on them through typed attributes.



ACTIVITY DIAGRAM:

This diagram models the control flow of one screening from start to finish, split into three swimlanes to show which actor owns each activity. After the Health Worker logs in and administers the EPDS, control passes to the System, which calculates the score and evaluates two decision nodes in sequence, one checking whether Item-10 is positive and one checking whether the score meets the cutoff. A positive Item-10 branches control to the Facility Supervisor lane for a same-day acknowledgment before merging back into the main flow. Both decision branches converge at the Determine Risk Tier, which is a merge point. After the referral is generated and viewed, I use a fork node to split the flow into two concurrent activities, since the Health Worker's follow-up update and the System's offline save do not depend on each other and can happen independently. A join node later synchronizes these two parallel flows back into a single path before the activity reaches its final node.

Component diagram

This diagram shows the technical components of my system and their dependencies. My Mobile Client UI has a dependency relationship with four components, Screening API, Auth API, Referral API and Escalation Notifier, meaning the UI relies on their interfaces but is not tightly coupled to their implementation. Each of these has a further dependency on a lower-level component, Screening Component, Security Component or the Sync and Backend Adaptor, which realize the actual business logic. Screening Component and Security Component each have a dependency on their own datastore, the Screening Store and Credential Store, both encrypted SQLite containers. The Sync and Backend Adaptor is the only component with an external dependency, since it makes API requests to the Central Sync and Dashboard Backend, a separate software system outside my system boundary. The Escalation Notifier has a similar external dependency on an SMS and push notification gateway.

